23 It would take 120 minutes to fill a swimming pool using water from 5 taps.

(a) How many minutes will it take to fill the pool if only 3 of the taps are used?

..... minutes

(2)

(b) State one assumption you made in working out your answer to part (a).

.....

.....

(1)

(Total for Question 23 is 3 marks)

24 A plane travels at a speed of 213 miles per hour.

(a) Work out an estimate for the number of seconds the plane takes to travel 1 mile.

Round 213 mph to 200 mph.

1 hour = 60 × 60 = 3600 seconds.

1 mark
Convert hours to seconds

	Miles	Seconds	
÷200	200	3600	÷200
	1	18	

1 mark
Scale distance and time together

3600 ÷ 200 = 36 ÷ 2 = 18.

1 mark
Estimated time in seconds

18 seconds

(3)

(b) Is your answer to part (a) an underestimate or an overestimate?
Give a reason for your answer.

Overestimate: the speed was rounded down,

so travelling the same mile takes longer.

(1)

1 mark
Link the lower speed to a greater time

(Total for Question 24 is 4 marks)

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